

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CL)

May 2014

CL403 Design of Structures-I

Date: 22.05.2014, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.
5. IS 456:2000, IS 800:2007, SP 16, and steel table are permitted.

SECTION - I

- Q - 1 Design and detail a beam of 5 m effective span, carrying a uniform load of 20 kN/m, [07]
if the compression flange is laterally unsupported. Assume $f_y = 250$ MPa. Give any two checks.
- Q - 2 (a) Design and detail a double angle tension member connected on each side of a 12 mm [07]
thick gusset plate, to carry an axial factored load of 300 kN. Use bolted connection.
The length of the member is 2.8 m.
- Q - 2 (b) Axial compression in a rafter of a truss is 80 kN. The spacing of the purlins is at 1.3 m [07]
and all the purlins are on the truss joints. Design the member with welded connections.

OR

- Q - 2 (a) Design and detail a suitable angle section to carry tensile force of 180 kN. Use welded [07]
connections. Take length of the member as 2 m.
- Q - 2 (b) Design and detail a double angle strut to carry an axial compressive factored load of [07]
220 kN. The length of the strut is 2.7 m. Welded connections are to be used to
connect the strut to 10 mm thick gusset plate.
- Q - 3 (a) Answer the following questions:
1. Show by means of sketch what you understand by the term "tacking bolts" with [01]
reference to bolted connections.
 2. Enlist two advantages of bolted connections over welded connections. [02]
 3. Differentiate 'Lap joint' and 'Butt joint' with reference to bolted connections by [04]
means of relevant sketches.
- Q - 3 (b) Answer the following questions:
1. Explain the following terms with respect to 'fillet weld' by means of sketches: [03]
(i) Size of weld, (ii) Effective throat thickness, (iii) Effective length

2. A diagonal member of the truss is an ISA 65 mm x 65 mm x 6 mm, welded in field [04]
to a gusset plate 8 mm thick. The grade of steel used for angle is Fe 410 grade
steel. Design the joint to the full strength of angle.

- If field weld is provided along the length of the member
- If field weld is provided along all sides of the angle

Take area of angle = 744 mm^2 and distance of c.g. from its heel is 18.1 mm.

OR

Q - 3 (a) Answer the following questions:

1. Show by means of sketch, what you understand by the following terms with [03]
reference to bolted connections:

(i) Gauge distance, (ii) Pitch of the bolts, (iii) Edge distance

2. A member of a steel roof truss consists of two angles ISA $75 \times 75 \times 6$ mm placed back [04]
to back on either side of 8 mm thick gusset plate. The member carries an ultimate
tensile load of 140 kN. Determine the number of 16 mm diameter 4.6 grade ordinary
bolts required for the joint. Assume f_u of plate as 410 MPa.

Q - 3 (b) Answer the following questions:

1. Enlist two advantages of welded connections over bolted connections. [02]
2. Differentiate 'Plug weld' and 'slot weld' by means of relevant sketches. [02]
3. Differentiate 'Fillet weld' and 'Butt weld' by means of relevant sketches. [03]

SECTION - II

Q - 4 Answers the followings:

[07]

- 1 According to IS 456:2000, the maximum compressive stress in concrete for design
purpose is taken as?
- 2 What is the value of minimum reinforcement in a slab?
- 3 Why minimum shear reinforcement is required?
- 4 Why dowels are provided in footings?
- 5 For isolated column footing design where is the location of one way shear and two
way shear?
- 6 What is the maximum and minimum % of longitudinal reinforcement provided in RC
column?
- 7 Minimum grade of concrete to be used for RCC work is _____.

Q - 5 Attempt any Two:

[14]

- 1 Design with detailing of a reinforced concrete beam supported on two walls 500 mm
thick, spaced at a clear distance of 6 m. The beam carries a super-imposed load of 30
kN/m. The size of the beam is restricted to 300 mm x 500 mm. Use M20 grade

concrete and Fe415 steel. Check this beam for deflection and cracking.

- 2 A short RCC column is to carry a factored load of 1900 kN. If the column is to be square, design the column. Assume $e_{min} < 0.05 D$. The materials are M20 grade concrete and Fe250 steel.
- 3 A RC beam 300 mm $\times$ 450 mm in cross-section is reinforced with 3 nos. 20 mm diameter bars, of grade Fe 250, with an effective cover of 50 mm. The ultimate shear at the section is 138 kN. Design the shear reinforcement.
 - (i) Using only vertical stirrups without bending any bar for resisting shear.
 - (ii) Bending 1 bar ϕ 20 mm at 45° to resist shear at the section.Use M20 grade concrete.

Q – 6(a) A square footing 3 m $\times$ 3 m and 600 mm thickness supports a 450 mm $\times$ 450 mm column. Design the dowels between the column and the footings if the characteristic loads at the base of column are 780 kN live load and 1000 kN dead load. Use M20 concrete and Fe 415 steel. [05]

(b) Attempt Any One:

[09]

- 1 A simply supported one-way slab of clear span 3.0 m is supported on masonry walls of thickness 350 mm. Slab is used for residential loads. Design the slab. The materials are M20 grade concrete and Fe415 steel. Live load shall be 2 kN/m². Give any three checks.
- 2 A drawing room of a residential building measures 4.3 m $\times$ 6.55 m. It is supported on 350 mm thick walls on all four sides. The slab is simply supported at edges with no provision to resist torsion at corners. Design and detail the slab using M 20 grade concrete and Fe 415 steel. Give any two checks.
